

TradeMirror — Analytics Indicator Reference

Complete inventory of every indicator, oscillator, and structural metric exposed by the centralized analytics engine (CandleV3 / SheetRow). This is the canonical set of inputs available to trend and momentum scoring logic.

PRICE

Raw OHLC data for the selected asset and timeframe. The foundation every other indicator is derived from.

- **Date / Timestamp:** Session date and ISO timestamp.
- **Open:** Session opening price.
- **High:** Session high.
- **Low:** Session low.
- **Close:** Session closing price — the primary input for every indicator that follows.
- **Symbol:** Instrument identifier (e.g. EURUSD).

SESSION CONTEXT

Bar-level derived metrics that describe the character of the current session and contextualize it against the prior bar.

- **% Change:** Session return from prior close to current close. Captures both the overnight gap and the intraday move. Positive = bullish session; negative = bearish.
- **Gap %:** Difference between today's open and yesterday's close, as a percentage. A positive gap signals overnight bullish sentiment; negative signals bearish. Gaps above 0.05% are meaningful for daily FX.
- **Range (pips):** $(\text{High} - \text{Low}) \times 10,000$. Total session volatility in pips. A wide range vs recent average indicates a high-conviction session; a narrow range signals consolidation or indecision.
- **Body Fill %:** $|\text{Close} - \text{Open}| \div (\text{High} - \text{Low}) \times 100$. What fraction of the total range was captured by the candle body. High fill = directional conviction; low fill = wick-dominated indecision.
- **Body (pips):** $|\text{Close} - \text{Open}| \times 10,000$ — absolute body size.
- **Upper Wick (pips):** $\text{High} - \max(\text{Open}, \text{Close})$, in pips — distance buyers pushed beyond the body and gave back.
- **Lower Wick (pips):** $\min(\text{Open}, \text{Close}) - \text{Low}$, in pips — distance sellers pushed beyond the body and gave back.
- **Inside Bar:** Today's high is below yesterday's high AND today's low is above yesterday's low — the full session is 'inside' the prior bar. Signals volatility contraction. Breakouts above the high or below the low are common entry triggers.

VOLUME

Participation metrics. Coloured by session direction so bullish vs bearish flow is visible at a glance.

- **Volume:** Raw session volume from OANDA. Plotted as green bars when close $\geq$ open (Volume Bull), red bars when close $<$ open (Volume Bear).
- **Vol SMA(20):** 20-session simple moving average of volume — the baseline for measuring whether participation is elevated or suppressed. Volume significantly above this line amplifies the significance of the concurrent price move; volume below it suggests low-conviction conditions.
- **Volume Ratio:** Current volume $\div$ Vol SMA(20). Ratios ≥ 1.5 mark high-volume sessions; < 0.7 marks consolidation/low participation. Ratio combined with bar direction yields the high-volume bull/bear surge classifications.

AVERAGE PRICE & RATE OF CHANGE

Short-horizon drift and rate metrics. Useful for distinguishing consistent pressure from one-bar spikes.

- **Average Price (HLC/3 SMA5):** 5-session simple moving average of typical price $(\text{High} + \text{Low} + \text{Close}) \div 3$. Unlike a close-only MA, it weights the full session range. Close above this average = buyers dominate; below = sellers hold the edge.
- **Rate of Change (5):** $(\text{Close} - \text{Close}[5]) \div \text{Close}[5] \times 100$. Percentage gain or loss over exactly one trading week. Positive = net bullish week; negative = net bearish. Accelerating ROC confirms momentum; decelerating ROC warns of fatigue. Extreme readings signal overextension.
- **Average Delta (5-bar):** Mean of $(\text{Close}[i] - \text{Close}[i-1]) \times 10,000$ over the last 5 bars. Positive = net buying drift; negative = net selling drift. Small avg delta with large ROC = one or two sessions drove the move; large avg delta = consistent pressure across all five sessions.
- **Momentum(10):** $(\text{Close} - \text{Close}[10]) \div \text{Close}[10] \times 100$. Unbounded percentage change vs 10-session prior close. Captures raw speed of the move — accelerating positive momentum confirms trend strength; decelerating warns of slowdown.

MOVING AVERAGES

Trend baselines at multiple horizons. The full EMA + SMA stack is available to scoring logic.

- **SMA(20):** 20-period simple moving average. Equivalent to the Bollinger midline.
- **SMA(50):** 50-period simple moving average. Mid-term mean.
- **SMA(200):** 200-period simple moving average — the long-term trend anchor. Price above = bull-market context; below = bear-market context.
- **EMA(9):** Fastest exponential MA — highly responsive to recent price action. Acts as dynamic short-term support in uptrends. A close below EMA9 in an uptrend is often the first warning of weakness.
- **EMA(20):** Responsive short-to-medium-term trend filter. Reacts faster than SMA20. Price above in an uptrend confirms short-term bullish structure.
- **EMA(50):** Primary medium-term trend reference. Faster than SMA50 — earlier signals on trend changes. Institutionally watched.
- **EMA(100):** Intermediate trend anchor between EMA50 and EMA200.
- **EMA(200):** Long-term trend anchor. Price above = bull-market context; below = bear. EMA Golden Cross (EMA50 above EMA200) and Death Cross (EMA50 below EMA200) are widely followed structural signals.
- **EMA Stack Alignment:** Derived state — $EMA9 > EMA20 > EMA50 > EMA200$ = full bullish stack; reverse = full bearish stack. Strongest trend confirmation when all four align.

MACD

Trend / momentum hybrid using two EMAs and a signal smoother. Standard 12 / 26 / 9 settings.

- **MACD Line:** $EMA(12) - EMA(26)$. Positive = short-term average above long-term (bullish); negative = bearish. Zero-line crossovers are trend-change signals (lag price).
- **Signal Line:** 9-period EMA of the MACD line. MACD crossing above signal = buy trigger; crossing below = sell trigger. Crossovers far from zero are more reliable than those near zero.
- **Histogram:** $MACD - Signal$, plotted as bars. Green when $MACD > signal$ (bullish momentum); red when $MACD < signal$ (bearish momentum). Growing bars = accelerating momentum; shrinking bars = slowing / potential reversal. Histogram turns *before* the lines cross — leading indicator of momentum shifts.

PIVOT POINTS

Classic floor-trader pivots derived from the prior session's H/L/C. Provide intraday support/resistance levels.

- **Pivot Point (PP):** $(\text{Prior H} + \text{Prior L} + \text{Prior C}) \div 3$. The session's central reference. Price above PP = bullish bias; below = bearish bias.
- **Resistance R1 / R2 / R3:** Progressively stronger resistance above PP. A close above any R level turns it into new support. R3 is rarely tested in a single session — reaching it signals an exceptionally strong bullish move.
- **Support S1 / S2 / S3:** Progressively stronger support below PP. A close below any S level turns it into new resistance. S3 violations are typically associated with high-volatility, trend-driven sessions.

BOLLINGER BANDS

Volatility-adjusted envelope around the 20-period SMA. Standard settings: $SMA(20) \pm 2\sigma$.

- **BB Upper:** $SMA(20) + 2 \times stdev(20)$. Touches / pierces signal overbought in ranges OR breakout strength in trends.
- **BB Middle (SMA20):** The mean-reversion target in ranging markets; dynamic support (uptrend) or resistance (downtrend) in trends.
- **BB Lower:** $SMA(20) - 2 \times stdev(20)$. Touches / pierces signal oversold in ranges OR breakdown strength in trends.
- **BB Width:** $(Upper - Lower) \times 10,000$ (pips). Widens in high-volatility regimes; contracts during squeezes.
- **%B:** $(Close - Lower) \div (Upper - Lower) \times 100$. 100% = at upper band, 50% = at midline, 0% = at lower band. Normalized location within the bands.
- **Historical Volatility (annualized):** Annualized standard deviation of log returns $\times \sqrt{252}$. Rising = active/trending market; falling = consolidation. Cross-check with BB Width to confirm whether compression is genuine.

KELTNER CHANNELS

ATR-based envelope around EMA(20). Less reactive than Bollinger Bands — useful for trend-riding rules.

- **Keltner Upper:** $EMA(20) + 2 \times ATR(14)$. A close above signals bullish breakout or overbought extreme. In strong trends, price can ride the upper band.
- **Keltner Middle:** $EMA(20)$. Acts as a dynamic magnet — in ranges price gravitates back to it; in trends it acts as dynamic support / resistance. The slope of the midline shows current trend direction.
- **Keltner Lower:** $EMA(20) - 2 \times ATR(14)$. A close below signals bearish breakdown or oversold extreme. In downtrends, price can ride the lower band.

SQUEEZE

BB vs Keltner compression detector — identifies volatility-contraction setups that often precede sharp directional moves.

- **Squeeze State:** ON when Bollinger Bands are fully inside Keltner Channels (BB Upper < KC Upper AND BB Lower > KC Lower). OFF otherwise.
- **Squeeze Duration:** Count of consecutive bars in the current squeeze state. Extended squeezes (5+ bars) often precede the largest expansion moves.
- **Squeeze Fired:** Transitional event — squeeze was ON last bar, OFF this bar. The highest-conviction entry window; direction confirmed by squeeze momentum.
- **Squeeze Momentum:** $Close - average\ of\ [20\text{-bar mid-range}, SMA(20)] \times 10,000$. Direction (sign) and slope (rising/falling) of momentum at the fire moment.
- **BB / KC Width Ratio:** Numeric measure of compression — < 1.0 = squeezing, ≥ 1.0 = expanded.

ATR

Volatility benchmark in price units. Used for stop sizing and volatility regime classification.

- **ATR(14):** Wilder-smoothed average of True Range over 14 bars. True Range = $\max(H-L, |H-PrevC|, |L-PrevC|)$. Expressed in pips for FX. Rising = expanding ranges, wider stops required; falling = compression, tighter stops viable.
- **ATR SMA(20):** 20-bar SMA of ATR(14) — the benchmark for 'normal' volatility. $ATR > 1.25 \times ATR\ SMA(20)$ = elevated-volatility regime (size down). $ATR < 0.75 \times ATR\ SMA(20)$ = compression (precursor to breakouts).
- **ATR Ratio:** $ATR(14) \div ATR\ SMA(20)$. Used by the regime detector to classify volatility as Expanding (> 1.10) / Neutral / Contracting (< 0.90).

RSI (9 & 14)

Relative Strength Index in two flavours — fast (RSI 9) for entry timing, standard (RSI 14) for higher-quality bias signals. Both use Wilder smoothing.

- **RSI(9):** 9-period RSI. Highly responsive — useful for spotting short-term exhaustion quickly, but prone to false signals in strong trends. Above 70 = overbought; below 30 = oversold.
- **RSI(14):** Standard 14-period RSI. Smoother and more reliable — fewer but higher-quality signals. Above 70 = overbought; below 30 = oversold. Readings here carry more weight than RSI(9).
- **RSI Trend:** Slope state of RSI(14) — UP / DOWN / FLAT — derived from a short SMA of RSI(14). Crossover between RSI and its trend line is an early momentum-shift signal.
- **RSI Divergence:** Flag — true when price makes a new high/low but RSI fails to confirm. Classic early reversal warning.
- **Stochastic RSI %K:** Stochastic applied to RSI(14) values rather than price, smoothed 3 periods. Oscillates 0–100. Far more aggressive than raw RSI. Above 80 = overbought; below 20 = oversold.
- **Stochastic RSI %D:** 3-period SMA of %K. Lags %K — %K crossing above %D = buy signal; crossing below = sell signal. Crossovers in extreme territory are more reliable.

MOMENTUM OSCILLATORS

Bounded and unbounded oscillators that capture the speed and extremity of recent moves.

- **CCI (Commodity Channel Index, 20):** $(\text{Typical Price} - \text{SMA TP}) \div (0.015 \times \text{mean deviation})$. Above +100 = overbought; below -100 = oversold; ± 100 to 0 = neutral. Leading indicator — turns before price and excels at identifying extremes and divergences. Above ± 200 = unsustainable extreme.
- **CCI MA(14):** 14-bar SMA of CCI. Crossovers between CCI and its MA often signal early momentum shifts before the raw line reaches an extreme.
- **Williams %R (14):** $((\text{HighestHigh} - \text{Close}) \div (\text{HighestHigh} - \text{LowestLow})) \times -100$ over a 14-bar window. Range: -100 (most oversold) to 0 (most overbought). Above -20 = overbought; below -80 = oversold. When %R stays near 0 during a rally it confirms bullish strength; when it fails to reach -20 on bounces in a downtrend, bearish momentum is dominant.
- **Momentum(10):** $(\text{Close} - \text{Close}[10]) \div \text{Close}[10] \times 100$ — unbounded percentage change. Captures raw speed; accelerating positive momentum confirms trend strength.

ADX / DIRECTIONAL MOVEMENT

Wilder's Directional Movement system. Measures trend strength and direction separately.

- **+DI(14):** Positive Directional Indicator. Strength of upward price movement. When $+DI > -DI$, bulls control directional movement. Rising $+DI$ = strengthening upside pressure.
- **-DI(14):** Negative Directional Indicator. Strength of downward price movement. When $-DI > +DI$, bears dominate. A crossover of $-DI$ above $+DI$ with ADX above 25 is a classic bearish trend entry signal.
- **ADX(14):** Average Directional Index. Wilder-smoothed Dx . Measures trend strength *regardless of direction* — rises in both up and down trends. < 20 = ranging (avoid trend signals); ≥ 25 = trending; ≥ 40 = strong trend. Rising ADX confirms a developing trend; falling ADX signals fading regardless of direction.

ICHIMOKU CLOUD

Multi-component trend, momentum and support/resistance system. All five components are exposed.

- **Tenkan (Conversion Line):** $(9\text{-period high} + \text{low}) \div 2$. Plotted at the current bar. Short-term momentum. A TK cross above Kijun is a bullish entry signal. Acts as near-term support in uptrends.
- **Kijun (Base Line):** $(26\text{-period high} + \text{low}) \div 2$. Plotted at the current bar. Medium-term trend anchor and the most important Ichimoku S/R level. Price above Kijun = bullish; below = bearish.
- **Senkou A (Leading Span A):** $(\text{Tenkan} + \text{Kijun}) \div 2$, plotted 26 bars FORWARD. Forms the faster cloud boundary. When $A > B$ the cloud is bullish.
- **Senkou B (Leading Span B):** $(52\text{-period high} + \text{low}) \div 2$, plotted 26 bars FORWARD. Slower, stronger cloud boundary. When $B > A$ the cloud is bearish. Thicker cloud = stronger S/R zone.
- **Chikou (Lagging Span):** Today's close plotted 26 bars BACK. Confirms trend by comparing current close to historical price. Above price from 26 bars ago = bullish; below = bearish.
- **Price Above Cloud:** Boolean flag — true when $\text{close} > \max(\text{Senkou A}, \text{Senkou B})$. Headline bullish-context flag for scoring.
- **Price Above Kijun:** Boolean flag — true when $\text{close} > \text{Kijun}$. Primary intermediate-trend flag.

MARKET STRUCTURE

Foundational intelligence layer — describes the market narrative independently of indicators.

- **Swing High (SH):** Local price peak confirmed by being the highest high within the surrounding N bars on each side (default $N = 3$). Marks potential resistance and defines the upper boundary of structure.
- **Swing Low (SL):** Local price trough confirmed by being the lowest low within the surrounding N bars on each side. Marks potential support and defines the lower boundary of structure.
- **HH / HL:** Higher High and Higher Low — sequential bullish structure. Two consecutive $HH+HL$ = bullish state.
- **LH / LL:** Lower High and Lower Low — sequential bearish structure. Two consecutive $LH+LL$ = bearish state.
- **BOS (Break of Structure):** Continuation event — new SH printed above the prior SH in an existing bullish trend (or new SL below prior SL in a bearish trend). Confirms the prevailing direction.
- **CHOCH (Change of Character):** Reversal event — new SH printed in what was a bearish trend (or new SL in a bullish trend). First structural signal of a regime flip.
- **Structure State:** Bullish / Bearish / Range — derived from the last two $SH+SL$ pairs.

REGIME DETECTION

Descriptive classifier — labels current conditions so scoring logic can weight signals appropriately.

- **Regime State:** Trending / Ranging / Compression. *Compression* when $BB\ Width < 0.75 \times 20\text{-bar avg BB Width}$. *Trending* when $ADX > 25$. Otherwise *Ranging*.
- **Volatility State:** Expanding / Neutral / Contracting. Based on $ATR(14)$ vs $ATR\ SMA(20)$ — Expanding $> 1.10\times$, Contracting $< 0.90\times$, else Neutral.
- **Use in scoring:** Trend-following signals carry higher weight in Trending regimes; mean-reversion signals carry higher weight in Ranging regimes; both are suppressed during Compression (breakout-watch only).

CANDLE CONTEXT

Three-tier pattern detector run over the most recent 12 candles.

- **Tier 1 — Multi-Candle Story:** Highest reliability. Patterns requiring a prior directional move plus a pivot/pause candle plus a follow-through confirmation. Includes Morning Star, Evening Star, Bullish/Bearish Engulfing, Piercing Line, Dark Cloud Cover.
- **Tier 2 — Single-Candle Rejection:** Single candles with extreme wicks. Context-dependent: Hammer (bullish at bottom of decline), Hanging Man (bearish after rally), Inverted Hammer (needs bullish confirmation), Shooting Star (needs bearish follow-through).
- **Tier 3 — Needs Confirmation:** Structural signals requiring the next candle to confirm — early alerts, not entry signals. Tweezers (defended level), Harami, Harami Cross.

FAILURE SWING

Statistical view of intra-candle rejection — quantifies how today's wick rejection compares to historical context.

- **Failure Swing Value:** On bullish bars ($close > open$) = $(Open - Low) \times 10,000$ — the lower wick. On bearish bars ($close < open$) = $(High - Open) \times 10,000$ — the upper wick. Doji bars are excluded.
- **Failure Swing CDF:** Cumulative distribution function over all historical qualifying bars. Reads as: 'X% of sessions had a failure swing $\leq$ this pip value'. Steep early rise = small failure swings dominate; gradual slope = wide distribution / frequent large wicks.
- **Percentile Bands (p25 / p50 / p75 / p90):** Reference levels on the CDF. The 75th percentile separates routine intraday noise from meaningful rejection. Today's value above p75 = above-average rejection; above p90 = extreme session; below p50 = quiet/low-conviction day.

DERIVED FLAGS & STATES

Boolean / categorical outputs already computed by the engine — ready for direct use in scoring rules without additional math.

- **Inside Bar:** Boolean — current bar fully contained in prior bar's range.
- **Price Above Cloud:** Boolean — close above max(Senkou A, Senkou B).
- **Price Above Kijun:** Boolean — close above Kijun line.
- **RSI Divergence:** Boolean — price/RSI divergence detected.
- **RSI Trend:** Categorical — UP / DOWN / FLAT, slope of RSI(14).
- **Squeeze State:** Boolean — BB inside KC.
- **Structure State:** Categorical — Bullish / Bearish / Range.
- **Regime State:** Categorical — Trending / Ranging / Compression.
- **Volatility State:** Categorical — Expanding / Neutral / Contracting.
- **Last BOS / CHOCH:** Most recent structural event with direction (bull / bear).

APPENDIX — Suggested Scoring Groupings

A starting partition for trend & momentum scoring. Each group is independent and can carry its own weight in the composite score. Indicators may participate in more than one group if their interpretation differs (e.g. MACD line = trend, MACD histogram = momentum).

Trend (directional bias)

- EMA stack alignment (9 / 20 / 50 / 100 / 200)
- SMA(50), SMA(200) — long-horizon anchor
- MACD line vs zero, MACD signal cross
- ADX > 25 with +DI vs –DI dominance
- Price vs Kijun, Price vs Ichimoku Cloud
- Market Structure state + most recent BOS / CHOCH
- Average Price (HLC/3 SMA5) vs close

Momentum (rate / extremity)

- RSI(9), RSI(14), RSI Trend, RSI Divergence
- Stochastic RSI %K / %D and their crossover
- MACD histogram (expanding vs contracting)
- CCI vs ± 100 , CCI vs CCI MA(14)
- Williams %R vs –20 / –80
- Momentum(10), Rate of Change(5)
- Average Delta (5-bar) — consistency of pressure

Volatility / regime weight

- ATR(14) vs ATR SMA(20) — Expanding / Neutral / Contracting
- Bollinger Width and %B
- Historical Volatility (annualized)
- Squeeze state and duration
- Regime state — Trending / Ranging / Compression

Confirmation / filters

- Volume ratio vs Vol SMA(20)
- Body Fill %, Range (pips), Inside Bar
- Failure Swing percentile rank (today vs p25 / p50 / p75 / p90)
- Candle Context tier (T1 / T2 / T3) and bias
- Pivot Point context — distance to R1/R2/R3 and S1/S2/S3